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Task 1: Case Study

Problem Statement

Choose a real-world case study such as a **KL ERP System**, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World Case Study

KL ERP System

1. Functional Requirements

Student

- Register and Login
- Academic Course Registration
- View Course Materials
- View Attendance
- View Internal Marks
- View Timetable
- Take Online Exams
- View Examination Results
- Pay Academic Fees
- Access Hostel Services
- Access Library Services
- View Transportation Details
- Submit Feedback
- View Student Profile

Faculty

- Login
- Manage Course Materials
- Record Student Attendance
- Upload Internal Marks

- Conduct Online Exams
- Publish Examination Results

Administrator

- Manage Student Records
- Manage Faculty Records
- Manage Course Registration
- Manage Timetable
- Manage Examination Details
- Generate Academic Reports

Finance Office

- Verify Fee Payments
- Update Student Fee Records
- Generate Fee Reports

Librarian

- Issue Books
- Receive Returned Books
- Manage Library Records

2. User Stories

User ID	User Story	Priority
US-01	As a student, I want to register so that I can access the KL ERP System.	High
US-02	As a student, I want to log in securely so that I can access my academic information.	High
US-03	As a student, I want to register for courses online so that I can select my subjects.	High
US-04	As a student, I want to view my attendance so that I can monitor my academic progress.	High
US-05	As a student, I want to view internal marks so that I can evaluate my performance.	Medium
US-06	As a student, I want to view my timetable so that I can attend classes on time.	High

US-07	As a student, I want to pay fees online so that I can complete payments conveniently.	High
US-08	As a student, I want to view examination results so that I can know my academic performance.	High
US-09	As a faculty member, I want to record attendance so that student attendance is updated accurately.	High
US-10	As a faculty member, I want to upload internal marks so that students can view their scores.	High
US-11	As an administrator, I want to manage student records so that academic data remains accurate.	Medium
US-12	As an administrator, I want to generate academic reports so that institutional performance can be monitored.	Medium

3. Selected Software Development Process Model

Agile Scrum

Reason for Selection

- Requirements may change frequently.
- Continuous feedback from students and faculty.
- Incremental feature delivery.
- Faster development cycles.
- Easy bug fixing and maintenance.
- Suitable for web-based ERP applications.

Sprint Plan

Sprint	**Features**
Sprint 1	Student Registration & Login
Sprint 2	Course Registration & Timetable
Sprint 3	Attendance & Internal Marks
Sprint 4	Examination & Results
Sprint 5	Fee Payment Module
Sprint 6	Library & Hostel Management
Sprint 7	Testing & Deployment

4. Programming Language Selection

Backend

Java (Spring Boot)

Reasons

- Secure
- Scalable
- Enterprise-level applications
- REST API support
- Excellent database connectivity

Frontend

- HTML5
- CSS3
- JavaScript
- React.js

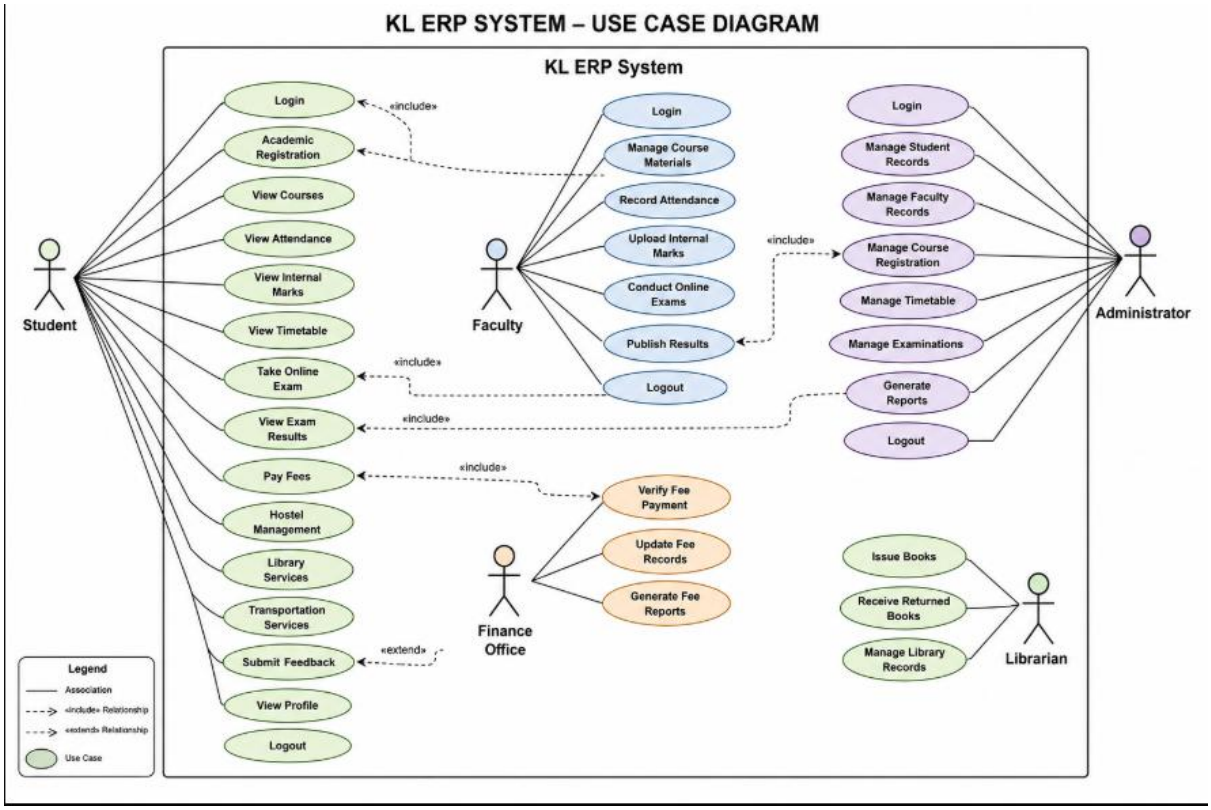
Database

- MySQL

Other Technologies

- REST API
- JWT Authentication
- Git
- Docker
- AWS Cloud

5. System Architecture



6. Implementation Strategy

Phase 1 – Requirement Analysis

- Gather requirements from students, faculty, administrators, finance office, and library staff.
- Create Software Requirement Specification (SRS).
- Identify functional and non-functional requirements.
- Prepare use case diagrams.

Phase 2 – System Design

- Design database schema.
- Design user interface (UI) wireframes.
- Create UML diagrams.
- Develop system architecture.

Phase 3 – Development

- Implement user authentication.
- Develop student registration and course registration modules.

- Develop attendance and examination modules.
- Implement fee payment module.
- Develop library and hostel management modules.
- Build administrator dashboard.
- Integrate notification and report generation features.

Phase 4 – Testing

Unit Testing

- Login
- Student Registration
- Course Registration
- Attendance Module
- Fee Payment

Integration Testing

- Course Registration + Timetable
- Examination + Results
- Fee Payment + Student Records

System Testing

- Complete KL ERP System testing

User Acceptance Testing (UAT)

- Verify academic and administrative requirements.

Phase 5 – Deployment

- Docker Container
- AWS EC2
- MySQL Database
- Nginx Web Server

7. Evaluation Strategy

Evaluation Parameter	Method
Functional Accuracy	Requirement verification

Evaluation Parameter	Method
Performance	Load testing using JMeter
Security	Authentication and penetration testing
Reliability	Continuous monitoring
Usability	User feedback survey
Scalability	Concurrent user testing
Maintainability	Code review and documentation

8. Expected Outcomes

- Secure and centralized ERP platform.
- Efficient student course registration.
- Accurate attendance and examination management.
- Reliable fee payment processing.
- Improved communication between students, faculty, and administration.
- Reduced paperwork and manual processing.
- Enhanced academic record management.
- Scalable architecture for future enhancements.